VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

PHYSICS 9478/01

Multiple-Choice Questions **22 September 2026**

1 hour

Candidates answer on OMS.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST
Write your name, class and subject on the OMS.
Write your NRIC number clearly and shade the OMS accordingly.

The use of an approved scientific calculator is expected, where appropriate.

There are **30** questions in this paper. Answer **all** questions. For each question, there are four answers **A**, **B**, **C** and **D**. Shade your answers on the OMS. Use only 2B pencil. You are advised to spend 40 minutes on this section.

This document consists of **17** printed pages.

Data

speed of light in free space	$c = 3.00 \times 10^{8}$ m s^{-1}
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} $(1 / (36\pi)) \times 10^{-9}$ F m^{-1}
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = F / A$
gravitational potential	$\phi = -Gm / r$
temperature	$T / \text{K} = T / ^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal molecule	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
capacitors in series	$1/C = 1/C_1 + 1/C_2 + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2$
charging a capacitor	$Q = Q_0 \left[1 - e^{-\frac{t}{\tau}} \right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2} n^2$
radioactive decay	$x = x_0 \exp(-\lambda t)$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

1 The energy E stored in a certain electronic component is given by

$$E = \frac{Q^n}{2k}$$

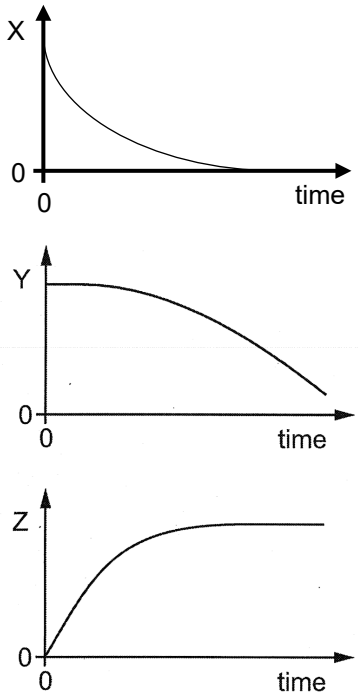
where Q is the total charge in the component, n is an unknown integer and k is a physical quantity with SI base units $A^2\,s^4\,kg^{-1}\,m^{-2}$.

What is the value of n?

A -2 **B** -1 **C** 1 **D** 2

2 An object is dropped at time t = 0 s from a tall building. Air resistance is significant.
Three graphs are plotted against time:

- the height of the object above the ground
- the speed of the object
- the magnitude of the resultant force on the object.



What are the quantities X, Y and Z?

	height of the object above the ground	speed of the object	magnitude of the resultant force on the object
A	X	Y	Z
B	X	Z	Y
C	Z	Y	X
D	Y	Z	X

3 Which person in the following situations is experiencing weightlessness?

A An athlete clearing the bar in a high jump.
B A swimmer floating in water.
C A parachutist falling at terminal speed.
D A pilot in a hovering helicopter.

4 Which of the following statements is true?

A When a moving object experiences friction, the work done against friction is independent of the path taken by the object.

B An object which experiences a resistive force always reaches terminal speed.

C Friction always hinders motion.

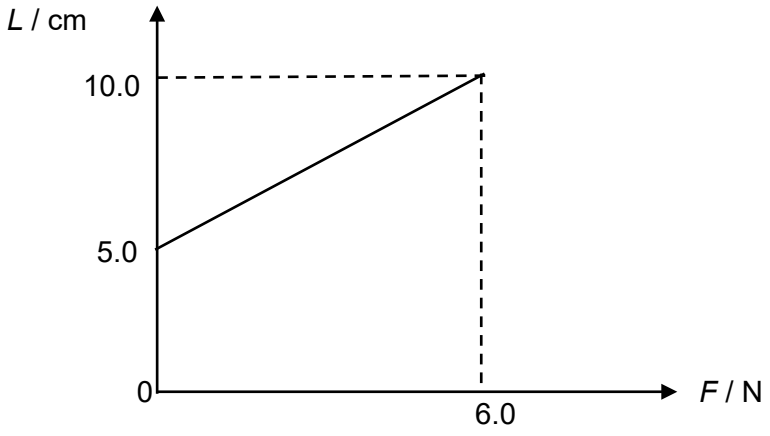
D There are certain situations in which friction does not cause energy to be lost to the surroundings.

5 A man pulls a rope attached to a sled. The rope makes an angle of 30° with the horizontal. The man exerts a force of 20.0 N on the rope to move the sled horizontally through a distance of 40 m.

What is the work done by the man?

A 400 J **B** 490 J **C** 690 J **D** 800 J

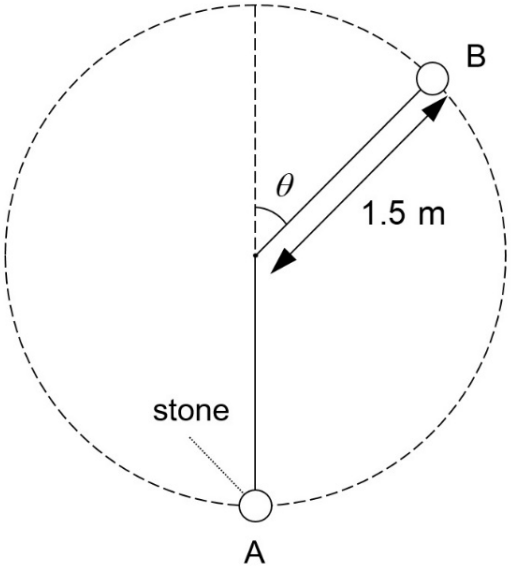
6 The graph below shows how the length L of a spring suspended from one end varies with the force F exerted on the other end.



Initially the spring has an extension of 2.5 cm. What is the work done to extend the spring by a further 2.5 cm?

A 0.11 J **B** 0.15 J **C** 0.26 J **D** 0.45 J

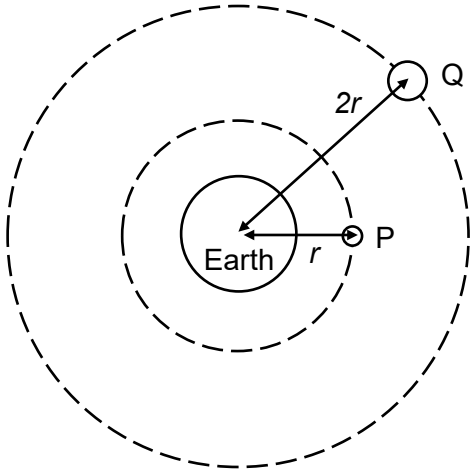
7 A stone attached to a light inextensible string at position A is given an initial push to the right. It subsequently moves in a vertical circle of radius 1.5 m and the string just slackens at position B when it makes an angle of θ to the vertical.



Given that the velocity of the stone at position B is 2.2 m s^{-1}, what is θ?

A 9° **B** 19° **C** 71° **D** 81°

8 Two satellites P and Q are in circular orbits about the Earth. Satellite P has mass *m* and it is in an orbit of radius *r*. Satellite Q has mass 2*m* and it is in an orbit of radius 2*r*.



Which of the following statements is true?

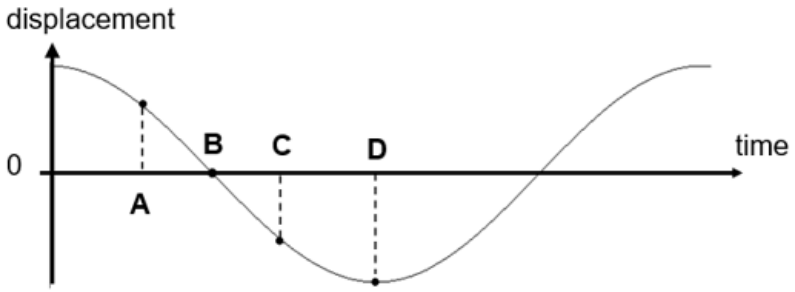
A The speed of satellite Q is greater than that of satellite P.

B The speed of satellite Q is the same as that of satellite P.

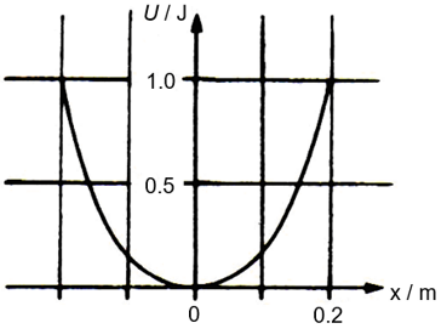
C The kinetic energy of satellite Q is greater than that of satellite P.

D The kinetic energy of satellite Q is same as that of satellite P.

9 The diagram shows the graph of displacement against time for a body performing simple harmonic motion. At which time, **A, B, C** or **D** is the body travelling and accelerating in the opposite direction?



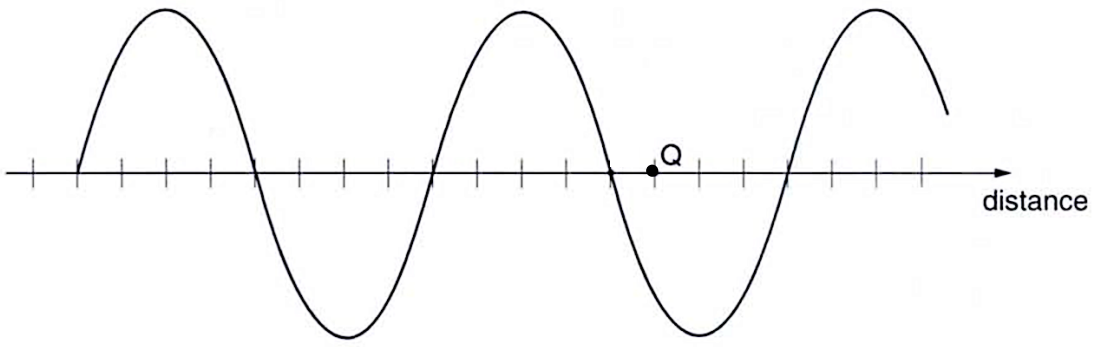
10 A particle of mass 4.0 kg moves with simple harmonic motion and its potential energy U varies with position x as shown in the figure below.



What is the period of oscillation of the particle?

A 0.25 s **B** 0.89 s **C** 1.8 s **D** 2.5 s

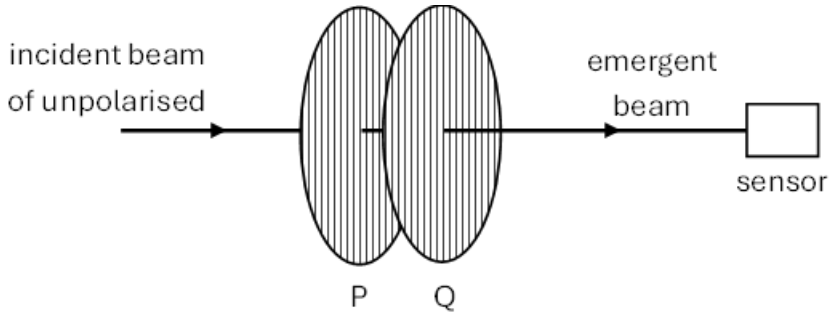
11 The diagram shows the displacement of a transverse wave along a stretched cord at a particular instant. The wave travels from left to right with frequency 20 Hz.



What is the shortest time before the displacement at Q becomes zero?

A 6.25 ms
B 18.8 ms
C 31.3 ms
D 43.8 ms

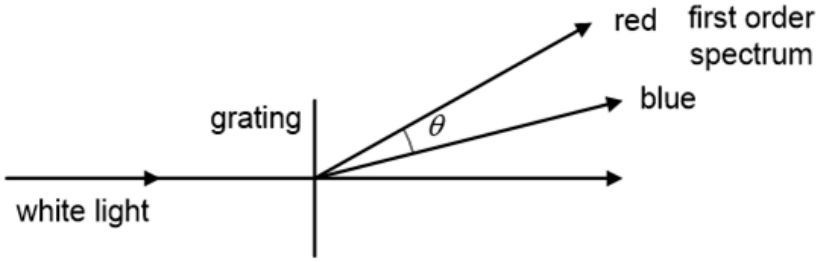
12 Two polarising filters, P and Q, are placed in front of a light sensor. Their transmission axes are initially parallel. Unpolarised light passes through them and the sensor records an intensity I.



Filter Q is then rotated while filter P remains fixed.
Which angle through which Q may be rotated would reduce the detected intensity by 36%?

A 23º **B** 37º **C** 50º **D** 53º

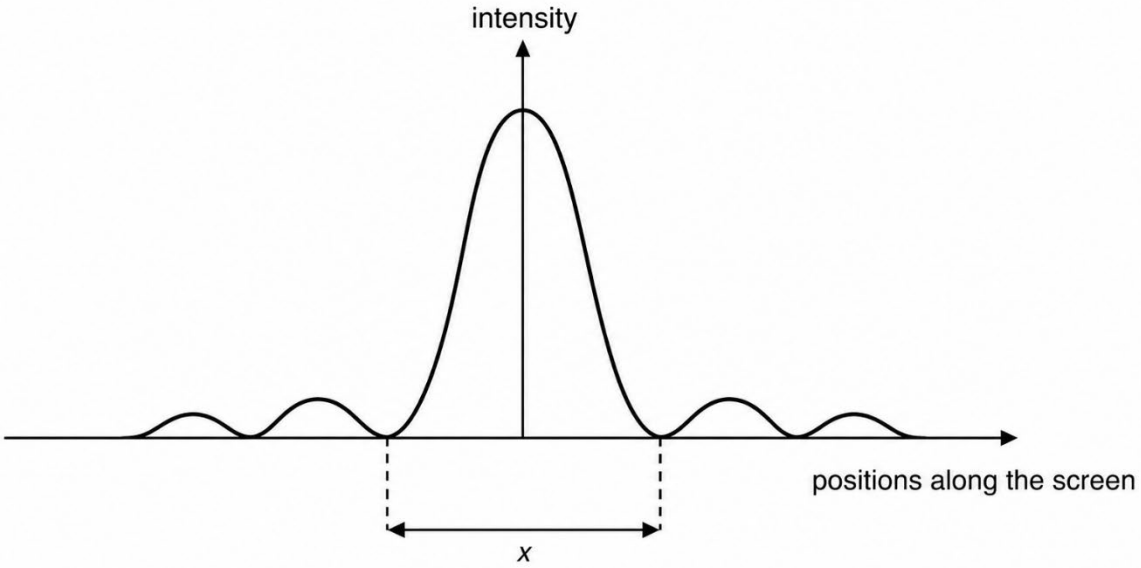
13 A compact spectrometer uses a diffraction grating containing 4.0×10^5 lines per metre. White light with wavelengths from 420 nm to 680 nm is incident normally on the grating.



What is the angular separation between the blue and red ends of the first-order spectrum?

A 6.1º **B** 9.7º **C** 15.8º **D** 25.5º

14 A parallel beam of monochromatic light passes through a narrow slit and forms a diffraction pattern on a distant screen. The width x is the distance between the first minima on either side of the central maximum.



Which statement is **incorrect**?

A Moving the screen closer to the slit increases x.

B Increasing the slit width generally allows more light to pass through the slit.

C Increasing the slit width decreases x.

D Increasing the wavelength increases x.

15 Heat is added to a pure substance in a closed container at a constant rate. A graph of the temperature of the substance as a function of time is shown below.



If L_f is the latent heat of fusion and L_v is the latent heat of vaporization of the substance, what is the value of the ratio $\frac{L_v}{L_f}$ for this substance?

A 1.0 **B** 2.4 **C** 3.5 **D** 5.0

16 An ideal gas is held in a container of volume *V* at pressure *P*. The root-mean-square (r.m.s) speed of a gas molecule under these conditions is *u*. If the volume and pressure are now changed to *2V* and *2P* respectively, what will be the r.m.s speed of a molecule?

A u **B** $\sqrt{2}u$ **C** $2u$ **D** $4u$

17 The first law of thermodynamics may be expressed as

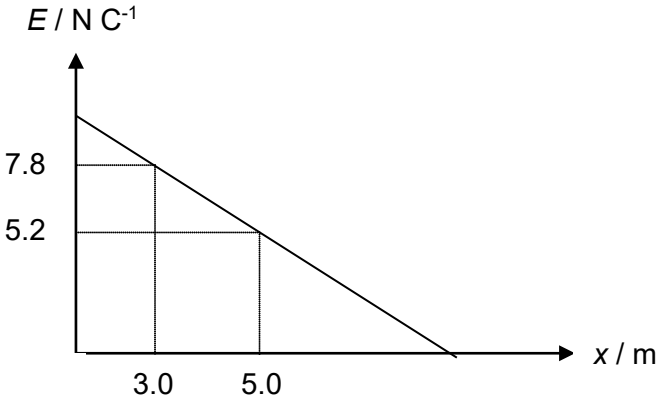
$$\Delta U = Q + W$$

where ΔU is the increase in energy of the system, Q is the thermal energy supplied to the system and W is the work done on the system.

Which of the following is correct for the case of an isothermal expansion of an ideal gas?

	ΔU	Q	W
A	zero	positive	negative
B	positive	zero	positive
C	zero	negative	positive
D	negative	zero	negative

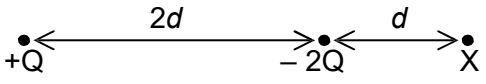
18 The graph below shows how the electric field strength E varies with displacement x from a point A. What is the change in potential for an electron if it is moved from a point 3.0 m away from A to another point 5.0 m away from A?



A −13 V **B** −1.3 V **C** 1.3 V **D** 13 V

19 Two charges $+Q$ and $-2Q$ are placed at a distance $2d$ apart. Point X is at a distance d away from the $-2Q$ charge as shown below:

11



What is the electrical potential energy of a $+3Q$ charge placed at X?

A $\dfrac{5Q^2}{4\pi\varepsilon_0 d}$ **B** $\dfrac{-5Q^2}{4\pi\varepsilon_0 d}$ **C** $\dfrac{-Q^2}{8\pi\varepsilon_0 d^2}$ **D** $\dfrac{Q^2}{12\pi\varepsilon_0 d^2}$

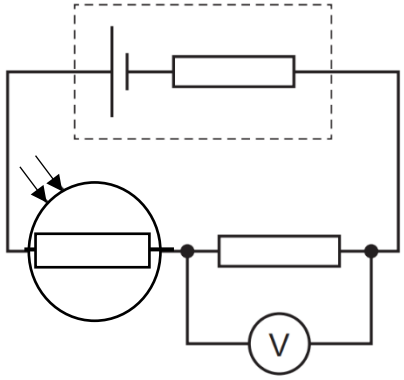
20 An electric current of 12 A flows in a wire of length 0.35 m and diameter 1.5 mm.

The average drift speed of the free electrons in the wire is 5.0×10^{-4} m s^{-1}.

How many free electrons are there in the wire?

A 1.3×10^{16} **B** 5.3×10^{22} **C** 1.5×10^{23} **D** 4.3×10^{23}

21 A cell with internal resistance is connected to a light-dependent resistor (LDR), a fixed resistor and a voltmeter as shown.



The voltmeter reading increases.

Which quantity **decreases** as the voltmeter reading increases?

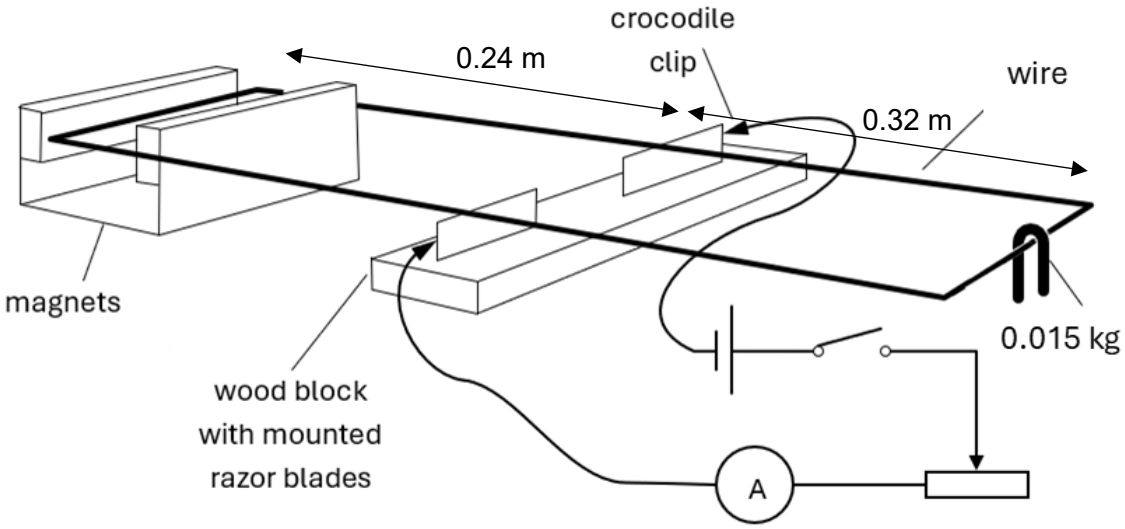
A the charge moving through the cell per unit time

B the energy transferred to the fixed resistor per unit charge

C the intensity of the light incident on the LDR

D the terminal potential difference across the cell

22 A laboratory current balance consists of a horizontal copper wire supported on two metal supports. A section of wire of length 0.18 m lies perpendicular to a uniform magnetic field produced by a magnet. When a current flows in the wire, a magnetic force due to the magnetic field acts on it.

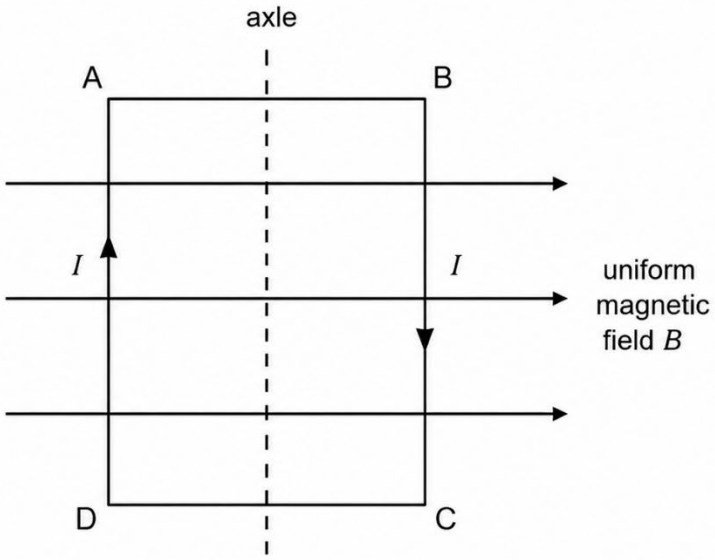


The magnetic force acts at a perpendicular distance of 0.24 m from the pivot. A hanging mass 0.015 kg, placed 0.32 m from the pivot, balances the wire when the switch is turned on and the current is 1.5 A.

What is the magnetic flux density produced by the magnet?

A 0.18 T **B** 0.48 T **C** 0.73 T **D** 0.95 T

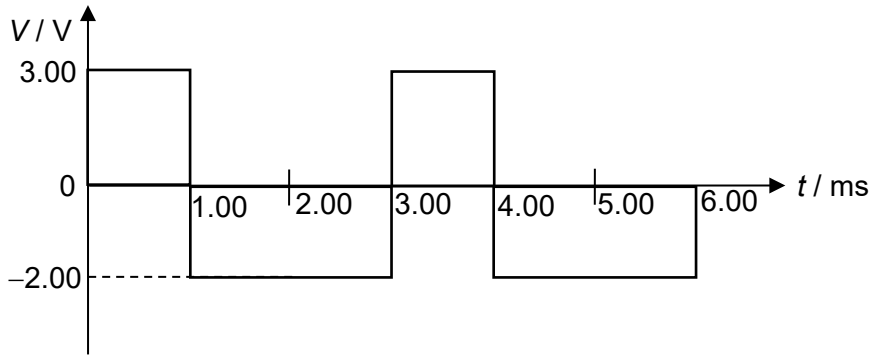
23 A rectangular coil ABCD is mounted on a vertical axle. Its plane is parallel to the field lines of a uniform horizontal magnetic field. Current flows upwards through side DA and downwards through side BC.



When the current is first switched on and before the coil moves, which statement is correct?

A There is no magnetic force on sides AB and CD.
B There is no magnetic force on sides BC and DA.
C Sides BC and DA experience forces in the same direction.
D All four sides experience magnetic forces of equal magnitude.

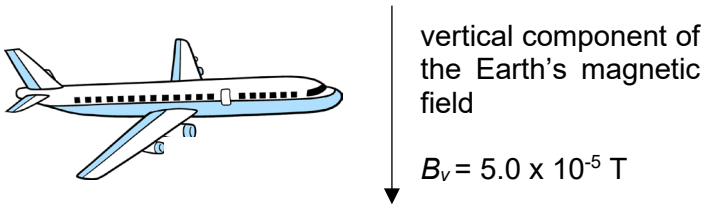
24 The graph below shows how the voltage of an a.c. source *V* varies with time *t*.



What is the r.m.s. voltage of the source?

A 2.12 V **B** 2.38 V **C** 2.50 V **D** 2.55 V

25 A plane heads East at a speed of 30 m s^{-1}, cutting the vertical component of the Earth's magnetic field of flux density B_v = 5.0 x 10^{-5} T acting downwards.



Taking the wingspan of the plane to be 17 m, what is the e.m.f generated across the wing and which side of the wing will be at higher potential?

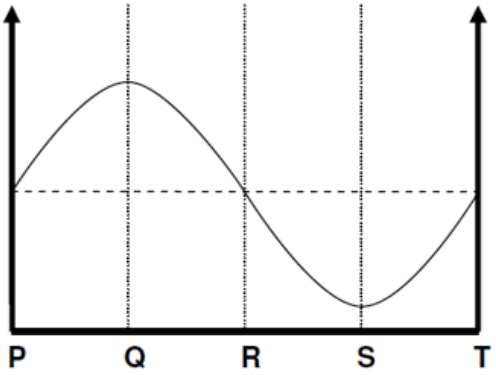
	E.m.f generated/mV	Wing which is at higher potential
A	0	None
B	0.030	Left
C	26	Right
D	26	Left

26 A plane circular coil of 30 turns, each of diameter 1.0 cm, rotates 1800 times each minute about a diameter which is perpendicular to a uniform magnetic field of flux density 4.0 $\times$ 10^{-5} T.

What is the maximum instantaneous value of the induced electromotive force?

A 1.2 $\times$ 10^{-6} V **B** 2.8 $\times$ 10^{-6} V **C** 1.8 $\times$ 10^{-5} V **D** 2.3 $\times$ 10^{-5} V

27 The wave function for an electron trapped in an infinite square well is shown in the figure below.



At which position(s) is the probability of locating the electron the highest?

A Q only

B Q and S only

C P, Q and R only

D P, R and T only

28 A proton has a kinetic energy of 1.00 MeV. If its momentum is measured with an uncertainty of 1.00 %, what is the minimum uncertainty in its position?

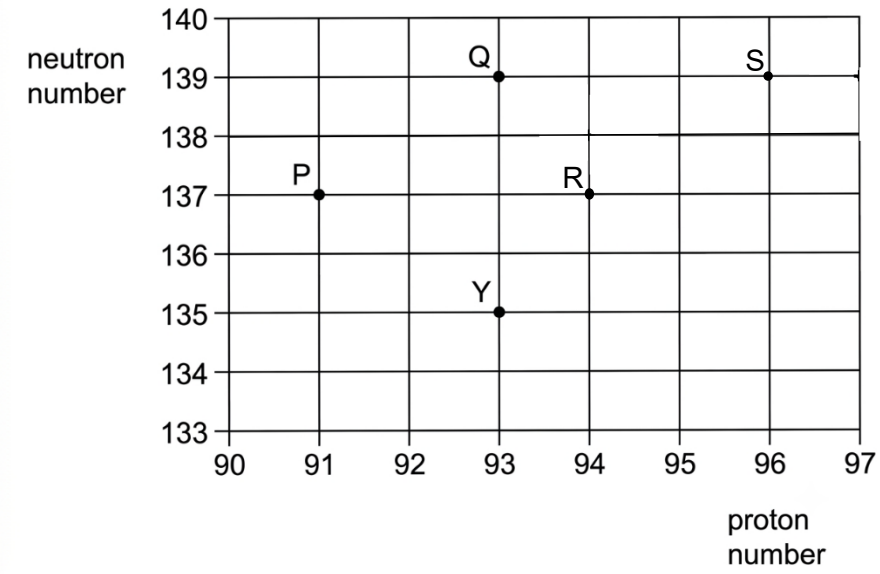
A 4.14×10^{-19} m

B 1.15×10^{-18} m

C 2.87×10^{-12} m

D 1.99×10^{-11} m

29 Nuclide Y is formed after two successive decays from another nuclide. Each decay can either be an alpha-emission or a beta-emission. Four other nuclides, P, Q, R and S are shown on the diagram.



Which of the nuclides P, Q, R and S is the initial nuclide that, after two successive decays, becomes Y?

A P **B** Q **C** R **D** S

30 A detector of ionising radiation gives a background count rate of 24 per minute.

A radioactive source is placed close to the detector and the reading is 532 counts per minute. What will be the reading after two half-lives of the source?

A 127

B 133

C 151

D 157